WORKED SOLUTION

STEP III 2024, Question 12

Probability and Statistics •

STEP III 2024, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

For a point chosen uniformly in the unit square, $\mathrm{P}(R \leq r)$ is just the area of the part of the square inside the circle of radius r about the origin. Once $r > 1$ the circle pokes out of two sides, and the region left inside the square splits neatly into two right-angled triangles and a sector. For the expectation, differentiate to get the density and integrate $rf(r)$; an integration by parts makes the awkward $\cos^{-1}$ term collapse, and a hyperbolic substitution finishes the job.

Solution

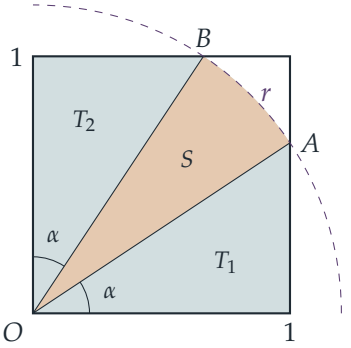
Part (i)

Since probability equals area, $\mathrm{P}(R \leq r)$ is the area of the set of points of the square at distance at most r from O.

When $0 \leq r \leq 1$ the quarter-disc $x, y \geq 0$, $x^2 + y^2 \leq r^2$ lies entirely inside the square, so

$$\mathrm{P}(R \leq r) = \tfrac{1}{4}\pi r^2 \qquad (0 \leq r \leq 1).$$

When $1 \leq r \leq \sqrt{2}$ the circle $x^2 + y^2 = r^2$ meets the side $x = 1$ at $A = (1, \sqrt{r^2 - 1})$ and the side $y = 1$ at $B = (\sqrt{r^2 - 1}, 1)$; both lie on the square because $0 \leq \sqrt{r^2 - 1} \leq 1$. Let α be the angle OA makes with the x-axis, so $\cos\alpha = 1/r$, i.e. $\alpha = \cos^{-1}(r^{-1})$. By symmetry in $y = x$, OB makes the same angle α with the y-axis.



The part of the square inside the circle is made up of:

- triangle T_1 with vertices O, $(1,0)$, A: right-angled at $(1,0)$, with legs 1 and $\sqrt{r^2-1}$, so area $\frac{1}{2}\sqrt{r^2-1}$;
- triangle T_2 with vertices O, $(0,1)$, B: the mirror image of T_1, also of area $\frac{1}{2}\sqrt{r^2-1}$;
- the sector S of radius r between OA and OB, whose angle is $\frac{\pi}{2}-2\alpha$, so its area is $\frac{1}{2}r^2\left(\frac{\pi}{2}-2\alpha\right) = \frac{1}{4}\pi r^2 - r^2\alpha$.

Adding,

$$\mathrm{P}(R \le r) = \sqrt{r^2-1} + \tfrac{1}{4}\pi r^2 - r^2\cos^{-1}(r^{-1}) \qquad (1 \le r \le \sqrt{2}),$$

as required. (At $r=1$ both formulas give $\frac{1}{4}\pi$, and at $r=\sqrt{2}$ we get $1+\frac{1}{2}\pi - 2\cdot\frac{\pi}{4} = 1$, as we should.)

Part (ii)

Write $F(r) = \mathrm{P}(R \le r)$ and $f = F'$ for the density. On $0 < r < 1$, $f(r) = \frac{1}{2}\pi r$. On $1 < r < \sqrt{2}$ we need

$$\frac{\mathrm{d}}{\mathrm{d}r}\cos^{-1}(r^{-1}) = -\frac{1}{\sqrt{1-r^{-2}}}\cdot\left(-\frac{1}{r^2}\right) = \frac{1}{r\sqrt{r^2-1}},$$

using $\sqrt{1-r^{-2}} = \sqrt{r^2-1}/r$ for $r>0$. Then, by the product rule,

$$\begin{aligned} f(r) &= \frac{r}{\sqrt{r^2-1}} + \tfrac{1}{2}\pi r - \left(2r\cos^{-1}(r^{-1}) + \frac{r^2}{r\sqrt{r^2-1}}\right) \\ &= \tfrac{1}{2}\pi r - 2r\cos^{-1}(r^{-1}). \end{aligned}$$

The two $\dfrac{r}{\sqrt{r^2-1}}$ terms cancel, which is a good sign. Hence

$$\begin{aligned} \mathrm{E}(R) &= \int_0^{\sqrt{2}} r f(r)\,\mathrm{d}r = \int_0^{\sqrt{2}} \tfrac{1}{2}\pi r^2\,\mathrm{d}r - 2\int_1^{\sqrt{2}} r^2\cos^{-1}(r^{-1})\,\mathrm{d}r \\ &= \frac{\pi(\sqrt{2})^3}{6} - 2I = \frac{\sqrt{2}\,\pi}{3} - 2I, \end{aligned}$$

where $I = \displaystyle\int_1^{\sqrt{2}} r^2 \cos^{-1}(r^{-1})\,\mathrm{d}r$.

Ask yourself How do we get rid of the inverse cosine inside I?

Integrate by parts, differentiating $\cos^{-1}(r^{-1})$ (whose derivative we found above is algebraic) and integrating r^2:

$$\begin{aligned} I &= \left[\tfrac{1}{3}r^3 \cos^{-1}(r^{-1})\right]_1^{\sqrt{2}} - \int_1^{\sqrt{2}} \frac{r^3}{3} \cdot \frac{1}{r\sqrt{r^2-1}}\,\mathrm{d}r \\ &= \tfrac{1}{3}\cdot 2\sqrt{2}\cdot\frac{\pi}{4} - 0 - \frac{1}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r, \end{aligned}$$

using $\cos^{-1}(1/\sqrt{2}) = \frac{\pi}{4}$ and $\cos^{-1} 1 = 0$. (The integrand is unbounded at $r = 1$, but it behaves like $1/\sqrt{2(r-1)}$ there, so the integral converges.) Therefore

$$\mathrm{E}(R) = \frac{\sqrt{2}\,\pi}{3} - \frac{\sqrt{2}\,\pi}{3} + \frac{2}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r = \frac{2}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r,$$

which is the required result.

Part (iii)

The $\sqrt{r^2-1}$ suggests $r = \cosh t$, with $\mathrm{d}r = \sinh t\,\mathrm{d}t$ and $\sqrt{r^2-1} = \sinh t$ (as $t \geq 0$). The limits become $t = 0$ and $t = T$, where $\cosh T = \sqrt{2}$, so $\sinh T = 1$ and

$$T = \cosh^{-1}\sqrt{2} = \ln\bigl(\sqrt{2} + \sqrt{2-1}\bigr) = \ln\bigl(\sqrt{2}+1\bigr).$$

Then

$$\begin{aligned} \int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r &= \int_0^T \frac{\cosh^2 t}{\sinh t}\,\sinh t\,\mathrm{d}t = \int_0^T \tfrac{1}{2}(1+\cosh 2t)\,\mathrm{d}t \\ &= \left[\tfrac{1}{2}t + \tfrac{1}{2}\sinh t\cosh t\right]_0^T = \tfrac{1}{2}\ln\bigl(\sqrt{2}+1\bigr) + \tfrac{1}{2}\cdot 1\cdot\sqrt{2}. \end{aligned}$$

Multiplying by $\frac{2}{3}$,

$$\mathrm{E}(R) = \tfrac{1}{3}\Bigl(\sqrt{2} + \ln\bigl(\sqrt{2}+1\bigr)\Bigr),$$

as required.

Checks

- F is continuous at $r = 1$ (both pieces give $\frac{\pi}{4} \approx 0.785$) and reaches 1 at $r = \sqrt{2}$, the far corner.
- The density $\frac{1}{2}\pi r - 2r\cos^{-1}(r^{-1})$ is zero at $r = \sqrt{2}$, as it must be: only the single corner point is that far away.

- Numerically, $\mathrm{E}(R) = \frac{1}{3}(1.41421 + 0.88137) \approx 0.7652$, sensibly between $\frac{1}{2}$ and $\sqrt{2}$. A computer simulation of 4×10^5 random points gave a mean distance of about 0.766, and its empirical distribution function agreed with F to within 0.002 at several values of r.

Common slips

- Taking the sector angle to be $\frac{\pi}{2} - \alpha$ instead of $\frac{\pi}{2} - 2\alpha$: there is a triangle cut off at *each* of the two sides.
- Getting the sign of the derivative of $\cos^{-1}(r^{-1})$ wrong: the two minus signs (from $\cos^{-1}$ and from r^{-1}) cancel, so it is positive.

Beyond STEP

This is a first taste of geometric probability: the mean distance from a corner of the unit square to a random point inside it. The companion problem, the mean distance between *two* random points in the unit square, has a similar but more elaborate closed form involving $\ln(1+\sqrt{2})$, and it is a classic exercise in double integrals and changes of variable. Grimmett and Welsh, *Probability: An Introduction*, covers distributions defined by geometry and how to compute expectations from distribution functions.

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